

SANTOSH THAPA

Mechanical Engineering Undergraduate | Biology Minor | Physics & Interdisciplinary Research
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PROFILE

Mechanical Engineering undergraduate with a minor in Biology and a broad foundation in life sciences, from cellular processes to whole-organism systems. Interested in fundamental physics, quantum phenomena, biomechanics, biological systems, and interdisciplinary research connecting engineering, physics, and biology. Seeking undergraduate research, engineering internship, and STEM opportunities.

EDUCATION

Midwestern State University B.S. Mechanical Engineering, Minor in Biology	Expected 2029
Dallas College Associate Degree	2026

TECHNICAL & SCIENTIFIC SKILLS

- Engineering mechanics & quantitative problem-solving
- Mathematical and physics-based analysis
- Mechanical systems & energy conversion
- Engineering design concepts & mechanical modeling
- Fundamental physics & quantum concepts
- Biology, physiology & biological systems
- Biomechanics & bioengineering concepts
- Scientific research & technical communication

SELECTED PROJECTS & ACADEMIC STUDIES

Automotive Engine Mechanics Study

- Studied the four-stroke engine cycle, piston motion, connecting rods, crankshaft, torque, power, and drivetrain motion.
- Applied physics and engineering principles to understand mechanical energy conversion.

Physics–Biology Interdisciplinary Study

- Explored how physical principles can describe biological movement, forces, structure, and function.
- Developed interest in biomechanics and the connection between mechanical engineering and living systems.

Engineering Design & Problem-Solving

- Applied mathematical and physics-based reasoning to conceptual mechanical engineering problems.
- Explored mechanical systems, design principles, modeling, and engineering problem-solving.

BIOLOGY BACKGROUND

Broad interest spanning cellular and molecular biology through whole-organism systems: Cell Biology • Genetics • Physiology • Anatomy • Botany • Zoology • Ecology • Living Systems • Structure–Function Relationships • Biomechanics • Bioengineering

LEADERSHIP & VOLUNTEERING

- Participated in environmental service and awareness events.
- Participated in engineering and mathematics STEM events.
- Held club leadership responsibilities during high school.
- Developed experience in teamwork, organization, event participation, and peer engagement.

RESEARCH INTERESTS

Quantum Physics • Fundamental Physics • Biomechanics • Mechanical Systems • Biological Systems • Bioengineering • Physics–Biology Interface

CAREER GOAL

Seeking opportunities in undergraduate research, engineering internships, STEM projects, and faculty-led research to develop practical engineering and research experience while exploring the connections between mechanical engineering, biology, and fundamental physics.